

Tissues

Definition :- A group of cells having similar origin, structure and function is called a tissue.

Types :- There are four types of basic tissues in the body.

1. Epithelial tissue
2. Connective tissue
3. Muscular tissue
4. Nervous tissue

Cartilage and bone are specialized connective tissues.

Epithelial Tissue :- It is a layer of cells, which covers the external surfaces (skin) or lines the internal surface of gastrointestinal, respiratory and urogenital tracts.

Functions :-

- Protection - It protects the body surface from drying or bacterial invasion.
- Transport - Mucous and particulate matters are carried to epithelial surfaces. Fluid may pass through the cell.
- Secretion - The cells secrete the product synthesized, either to the lumen or blood.
- Excretion - May excrete metabolic waste products.
- Absorption - It absorbs essential substances from the lumen of GIT and kidney tubules (where it is called reabsorption).
- Lubrication - Peritoneum, pleura and pericardial epithelium serve

this function.

- **Sensory:** In the skin (touch sensation), nasal mucosa (smell sensation) and tongue (taste sensation), it serves as sensory organ.

Types of EPITHELIUM.

Epithelium is classified into simple, pseudostratified and stratified varieties.

Simple epithelium:- Single layer of cells resting on a basement membrane.

- | | |
|--------------------|--|
| 1. Simple Squamous | <p>Irregular flat cells with height less than width.</p> <ul style="list-style-type: none"> • Distribution: Alveoli of the lungs. • Bowman's capsule & loop of Henle of kidney. • Endothelial cells lining blood vessels. |
| 2. Simple cuboidal | <p>Height and width are nearly equal and nuclei are central in position.</p> <ul style="list-style-type: none"> • Distribution: Thyroid follicles • Ducts of many glands. • Surface of the ovary (germinal epithelium) |
| 3. Simple Columnar | <p>Height of the cells is greater than width. Nuclei are elongated and placed towards the base.</p> <ul style="list-style-type: none"> • Distribution: Small bronchi and bronchioles, uterine tube. • Ependyma lining the cavities of the brain. • Efferent ductules of the testes. |

Non-ciliated Simple Columnar

Distribution: lining of gastrointestinal tract from stomach to rectum.

- Gall bladder (with brush border).

Stratified epithelium:- Epithelium is made up of many layers of cells. They are found in areas, which are subjected to friction.

1. Stratified squamous epithelium

Non-keratinised: Epithelium is made up of many layers of cells. They are found in ~~area~~ Basal cells resting on the basement membrane are columnar or cuboidal.

 - Superficial cells are squamous and flat, hence called stratified squamous epithelium.
 - Distribution: Epithelium lining mouth, pharynx and esophagus, anal canal, vagina and cornea.

Keratinised: The superficial layer consists of non-living cells with keratin in their cytoplasm. They are tough and water-resistant.

 - Distribution: Epidermis of the skin.
2. Stratified cuboidal epithelium

It consists of few layers of cuboidal cells.

Distribution: Ducts of sweat glands
3. Stratified columnar epithelium

It consists of two or more layers of cells. Basal cells are polyhedral, while superficial cells are columnar.

Distribution: Male urethra (membranous part).

Pseudostratified Columnar Epithelium.

Single layer of epithelial cells resting on a basement membrane.

- Some cells are shorter and do not reach the lumen, while tall cells reach the lumen.
- The nuclei of the cells therefore lie at different levels. This gives the impression of stratification (false stratification).

A. Pseudostratified non-ciliated

- Distribution: Male urethra (membranous and penile parts)
- Auditory tube
- Vas deferens

B. Pseudostratified ciliated

- Distribution: Upper part of the respiratory tract (trachea and larger bronchi).

Transitional Epithelium (Urothelium)

This is a stratified epithelium with three to four layers of cells. The deepest cells are columnar or cuboidal.

The middle layers are made up of polyhedral or pear-shaped cells. The cells of the surface are large and are shaped like an umbrella.

- Distribution: Renal pelvis, ureter, urinary bladder and proximal part of the urethra.

Transitional epithelium can be stretched considerably without being damaged. When stretched the cells become flattened.

Glandular Epithelium

The epithelial cells are specialized to perform secretory function. Such epithelial cell in-groups constitute glands (single

epithelial cell can also be a gland (unicellular).

There are two main types of glands.

1. Exocrine glands: When the secretion from the gland is poured through duct system, they are called exocrine glands, eg. salivary gland, sweat glands, Mammary glands.
2. Endocrine glands: These glands are 'ductless' and pour their secretion directly into the bloodstream. Their secretion is called 'hormone'. The cells of the endocrine glands are usually arranged in cords or in clumps with rich network of blood capillaries around them.
eg. pituitary gland, thyroid gland, adrenal gland

CONNECTIVE TISSUE

- Connective tissue has formed elements (fibers and cells) and amorphous substance (ground substance).
- This tissue binds various other tissues of the body.

Components of the connective tissue: Connective tissue has two components—cells and matrix, which has fibers and ground substance.

- Cells: Seven types of cells are found in the connective tissue.
- Fibers: The connective tissue fibers are of three types, Collagen, Elastic and reticular fibers.

Collagen Fibers

- In fresh state, they appear white and therefore are also called white fibers.
- It contains a protein called 'collagen', which is derived from tropocollagen.

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- Collagen fibers run in straight or wavy bundles. The individual fibers do not branch but bundles themselves can branch.
 - The fibers are soft but strong and flexible but they are not elastic.
 - Collagen fibers are formed and maintained by fibroblasts.
 - They are found in all types of connective tissue.
 - The individual collagen fibers are $1-12\ \mu\text{m}$ in diameter.
 - Depending upon their diameter, they are classified into four types:

Types and distribution.

- Type I: (Diameter of about $250\ \text{nm}$) skin, tendons and ligaments.
- Type II: (Diameter of about 20 to $100\ \text{nm}$) nose and ears.
- Type III: Blood vessels, muscles and organs holding them together.
- Type IV: Kidneys.

Reticular fibers.

- These are similar to collagen fibers (type-III) in chemical structure.
- They are thin fibers showing branching and hence form a network or a reticulum.
- They do not run in bundles.
- These fibers form the skeletal framework of the lymphatic organs.

Distribution.

- Basement membrane of the epithelial tissue.
- Walls of the blood vessels.

• Lymphatic organs.

Elastic fibers.

- Elastic fibers are thin and highly elastic.
- These fibers branch and ^{connection} anastomose with each other but do not form bundles.
- In bulk, they appear yellow, hence also called yellow elastic fibers.
- Elastic fibers can retain their structure after being stretched.

Distribution.

- Loose connective tissue
- Walls of the blood vessels
- Ligamentum flava
- Capsules of the glands.

Ground Substance (Intercellular substance)

The ground substance consists of water, carbohydrates, lipids and proteins. They together constitute matrix.

Cell Types.

1. Fibro blasts: Each cell is flattened or fusiform in shape with centrally placed nucleus. They are responsible for the formation of collagen, reticular and elastic fibers. They help in healing of wounds.
2. Macrophages (histiocytes or clasmatoocytes): They are irregular in shape. These cells are phagocytic in function. They ^{engulf} phagocytose bacteria and other foreign bodies. Their cytoplasm has hydrolytic enzymes.

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3. Plasma cells: The cells are rounded in shape with eccentrically placed nucleus. These cells produce antibodies to counteract the actions of antigens in defense mechanism of the body.
 4. Mast cells: Each cell is rounded in shape with a centrally placed nucleus. They secrete heparin, which is an anti-coagulant. They also produce histamine, which causes allergic reactions. responsible for inflammation
 5. Pigment cells: They are present in the iris and choroid of the eye and they produce colour.
 6. Reticular cells: They are present in the reticular connective tissue. They produce reticular fibres. They are also phagocytic in function.
 7. Fat cells (adipocytes): These cells are polygonal in shape with eccentrically placed nucleus. The cytoplasm contains large amount of fat.

Areolar Tissue

It is the most common connective tissue where collagen and elastic fibers are loosely arranged. The ground substance is semifluid in nature. They have plenty of fibroblasts and macrophages. They are transversed by nerves and vessels.

Distribution

- Subcutaneous tissue
- Submucous coat in the gastrointestinal tract
- Spaces between the organs.

Adipose Tissue.

- This is an aggregation of fat cells.
- It serves as insulating material to conserve body heat, as storage depot for food, as protective pad around organs.
- It is abundant in females (under the hormone influence) especially in pectoral region and gluteal region.

Distribution.

- Mesenteries and omenta
- Bone marrow
- Ischio rectal fossa, axilla, bony orbit.

Cartilage and Bone.

Cartilage

- Cartilage is a specialised dense connective tissue.
- It is hard but not rigid like bone.
- It can be bent and also brought back into its original form when bending force is withdrawn.
- This cartilage forms the 'skeletal' basis of some parts of the body (auricle of the ear, external nose).

General Features

- They are rigid structures, hence provide protection and support the organs. They can withstand the effects of pressure, pull or torsion.
- They are present in the body where elasticity and rigidity is required.
- They are avascular structures, nourished by diffusion from adjacent tissues.
- Repair of cartilages is slow and takes time due to its avascularity.
- There are three types of cartilage - hyaline, elastic and fibrocartilage.

Cartilage consists of:

1. Cells called chondrocytes
2. Fibers
3. Ground Substance.

1. HYALINE CARTILAGE.

- It is covered by a vascular fibrous membrane called 'perichondrium'. This perichondrium has an outer fibrous layer and inner cellular layer. The cellular layer consists of chondroblasts (immature chondrocytes).

- It has a homogenous matrix, which is glassy and transparent in appearance.
- The matrix contains collagen fibres (Type II) which run in parallel bundles.
- The chondrocytes are placed in 'lacunae' of the matrix.
- The ground substance is made up of carbohydrates and proteins.

Distribution

1. Costal cartilages of the ribs.
2. Cartilages of the larynx - thyroid and cricoid cartilage.
3. The tracheal rings.

2. ELASTIC CARTILAGE

- Structurally the elastic cartilage is mainly made up of elastic fibres and the chondrocytes.
- The surface of the elastic cartilage is covered by 'perichondrium'.
- Elastic cartilage is more flexible than hyaline cartilage.

Distribution.

- Pinna of the external ear.
- Epiglottis, Corniculate and cuneiform cartilages of the larynx.
- Medial part of the auditory tube.

3. FIBROCARILAGE (WHITE ~~CA~~ FIBROCARILAGE)

- Structurally the white fibrocartilage contains mainly thick bundles of collagen fibres (type I) and a few chondrocytes.
- It has no perichondrium.
- It is very tough and strong but resilient.

- It has great tensile strength and considerable elasticity.

Distribution.

- Glenoidal labrum of shoulder joint.
- Acetabular labrum of hip joint.
- Menisci of the knee joint.
- Intervertebral discs present between the bodies of vertebrae.

BONE.

- Bone is a highly vascular living connective tissue in which the matrix is calcified by the deposition of calcium phosphate.
- The human skeleton consists of 206 bones.

Functions of the bone.

- Bone provides supporting framework and shape for the body.
- Bone protects vital organs of the body (e.g. heart and brain).
- They help in transmission of the body weight.
- They provide attachment to the muscles and act as levers of the joints helping in locomotion.
- Bone is the storehouse of calcium salts.
- Involved in erythropoiesis.

Classification of the bone.

According to their position.

1. Axial: Bones forming the axis of the body: e.g. skull, ribs, vertebrae.
2. Appendicular: Bones of the limbs.

According to the shape.

1. Long bones: They have three parts: Upper end, lower end

and a middle shaft. The ends of these bones take part in forming the joint (articulates with other bone), e.g. bones of limbs (humerus, ulna, radius, femur, tibia, fibula).

2. Short bones: These bones are small and generally cuboidal in shape, e.g. carpal and tarsal bones.
3. Flat bones: These bones are expanded and are flat, e.g. sternum, scapula, ribs and parietal bone.
4. Irregular bones: The shape is irregular without any proper outline, e.g. vertebrae, sphenoid, temporal bones, etc.

According to Gross structures:

1. Compact (lamellar bone): Structurally it is made up of bony plates (lamellae) which are arranged compactly, e.g. outer cortical part of the long bone.
2. Spongy bone (cancellous): Structurally it is made up of bony plates, which are arranged irregularly leaving spaces in between them. It gives a spongy appearance, and e.g. flat bones, irregular bones and ends of the long bone.
3. Diploic bone: It consists of inner and outer tables of compact bone with an interval which is occupied by bone marrow and diploic veins, e.g. most of the cranial bones (parietal, frontal, occipital).

Special types of Bones.

1. Pneumatic bones: These are flat or irregular bones with hollow spaces in their body. These spaces contain air e.g. ethmoid, maxilla and mastoid part of temporal bone.
2. Sesamoid bones (sesamoid=seed): These bones develop within the tendon of some muscles.
 - They do not possess periosteum and Haversian system.
 - They ossify after birth.
 - They minimise the friction and also change the direction of the pull of a muscle.

For muscle example, pisiform bone (in the tendon of flexor carpi ulnaris muscle), patella (in the tendon of quadriceps femoris muscle), fabella (in the tendon of lateral head of gastrocnemius muscle).

Macroscopic Structure of a Bone.

- The long bone consists of two ends (epiphysis) and a shaft (diaphysis) ^{inside}.
- The shaft consists of a cylindrical cavity called 'medullary cavity', which is filled with bone marrow. The outer (cortical) part of the shaft is made up of compact bone.
- The two ends of the long bone are filled with tiny plates of bone containing numerous spaces. This is referred as 'spongy bone' to which the medullary cavity does not extend.
- The outer surface of the bone is covered by a highly vascular connective tissue membrane called 'periosteum' except at the articular surfaces. This articular surface is covered by articular cartilage, usually hyaline type.
- The medullary cavity is lined by another connective tissue membrane called 'endosteum'.

Bone Marrow.

It occurs in two forms, yellow marrow and red marrow. At birth the red marrow is present throughout the skeleton. After about fifth year of postnatal life, the red marrow is gradually ~~ch~~ replaced in the long bones by yellow marrow and only the required amount is there for blood production.

Microscopic Structure.

An adult long bone consists of following components:-

- Bone cells
- Matrix.

Compact Bone.

- The compact bone is made up of 'lamellae'.
- Lamellae are thin plates of bone consisting of collagen fibers embedded in ground substance.
- Lamellae are placed one over another.
- The spaces between the lamellae are called 'lacunae'.
- Lacunae are occupied by osteocytes.
- The adjacent lacunae are connected through canaliculi, which are occupied by cytoplasmic processes of osteocytes.
- Most of the lamellae are arranged in the form of concentric rings that surround a 'Haversian canal', which is present at the center of each ring.
- Haversian canals are placed parallel to medullary cavity and they are occupied by blood vessels and nerve fibers.
- Adjacent Haversian canals are connected by Volkmann's canal.

- One Haversian canal and lamellae around it constitute a Haversian system or an Osteon.

Matrix (Ground Substance)

The matrix of the bone consists of both organic and inorganic constituents.

- Organic constituent (25% of the matrix) is mainly made up of collagen fibers responsible for toughness and resilience of bone.
- Inorganic constituent (75% of the matrix) following mineral salts are calcium phosphate (85%), calcium carbonate (10%), small amount of calcium fluoride and magnesium phosphate.

BONE CELLS.

Osteoblasts.

- These are bone forming cells.
- They are more numerous in periosteum.
- The cells are ovoid, triangular or cuboidal in shape with oval nucleus.
- These cells are responsible for laying down the organic matrix of bone including the collagen fibers.
- They are responsible for calcification of the matrix.

Osteocytes

- These are mature bone cells.
- They are derived from osteoblasts after they have laid down the matrix.
- They are present in the 'lacunae' of the bone between the lamellae.

Osteoclasts.

- These are bone removing cells and found in relation to the surfaces of the bone.
- Osteoclasts are multinucleated large cells (diameter varies from 20 to 100 μm).
- The lysosomes present in their cytoplasm contain 'acid phosphatase'.
- Osteoclasts are involved in demineralization and removal of bone matrix.
- Osteoclasts are stimulated by parathyroid hormone.

Periosteum

- The external surface of the bone is covered by a connective tissue membrane called 'periosteum' except the articular surface.
- The periosteum consists of outer fibrous layer and inner cellular layer consisting of osteoblasts.

Ossification.

The process of bone formation is called 'ossification'. All the bones are developed from the mesenchymal tissue of the embryo. There are two types of ossification.

1. **Membranous ossification:** The embryonic mesenchymal tissue will directly form the bone, e.g. bones of the cranial vault, mandible, and clavicle.
2. **Cartilaginous ossification:** The mesenchymal tissue is first transformed into a 'cartilage'. Later this cartilage is ossified to form a bone.

Ossification of a long bone.

The ossification begins in one or more areas of future bone model. These areas are called centers of ossification.

Primary center of Ossification.

- The ossification starts in the central part of the cartilaginous model (i.e. at the center of the future shaft). The
- ~~part~~ The portion of the long bone developed from this primary center of ossification is called 'diaphysis'.
- The primary center of ossification normally appears before birth.

Secondary center of Ossification.

- These centers appear at the two ends of the long bone usually after birth.
- The portion of the long bone developed from secondary center of ossification is called 'epiphysis'.
- The two ends of the diaphysis, which are actively involved in growth, are called 'metaphyses'.
- In a long bone between epiphysis and diaphysis is a part of the cartilage remains unossified until epiphysis fuses with diaphysis and it is called 'epiphyseal plate'.
- They also undergo ossification at puberty.

★ Laws of Ossification.

1. The epiphysis which ossifies first (or appears first) unites (fuses) with the diaphysis last and the epiphysis which ossifies last fuses first. Exception: lower end of the fibula.
2. The end of a long bone where epiphysis appears first and fuses last is called the 'growing end' of the bone.
3. The direction of nutrient artery is always away from the growing end.
4. In the long bones, growing ends fuse with the shaft at about 20 years and the opposite ends at about

18 years.

Bone Development and Growth.

- The growth of the skeleton determines the size and proportions of our body.
- The bony skeleton begins to form about six weeks after fertilization.
- Bone growth continues through adolescence.

Factors Regulating the Bone Growth.

- Normal bone growth requires constant dietary source of calcium and phosphate salts.
- Vitamin A and C are essential for normal bone growth and remodelling. These vitamins are obtained from the diet.
- Vitamin D plays an important role in normal calcium metabolism by stimulating the absorption and transport of calcium and phosphate ions into the blood.
- The thyroid gland secretes the hormone calcitonin, which stimulates osteoblasts to produce new matrix.
- The secretion of parathyroid stimulates osteoclast activity.
- Growth hormone produced by the pituitary gland and thyroxine from the thyroid gland stimulate the bone growth.

Blood supply to a long Bone.

1. Nutrient artery
2. Epiphyseal arteries
3. Metaphyseal arteries
4. Periosteal arteries.

Nerve Supply to the Bone.

Bones are innervated by sensory nerves and injuries to the bone can be very painful.